American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
5	2/18/20	9:00 am	30 minutes	(•) Advisor ( ) Students	Claritta Khoury

Attendees:

Students: Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury. Advisor: Professor Ayman Kayssi

Briefly summarize the main discussions during the meeting:

The meeting focused on updates related to edema detection, hardware stabilization, signal processing improvements, and mobile app integration. The team discussed testing multiple strap and sensor configurations using the Arduino platform to measure limb circumference changes and classify edema severity (mild, moderate, severe) based on percentage variation. The approach includes implementing hysteresis logic to prevent unstable fluctuations when measurements return close to baseline values. The team also discussed establishing a personal baseline monitoring period of one to two weeks before triggering alerts, similar to wearable devices that learn individual physiological norms. Hardware discussions addressed voltage limitations when powering a 5V Arduino from a 3.7V Li-Po battery, highlighting the need for a reliable step-up converter and sourcing compact modules locally. The integration of a multiplexer was rdiscussed to expand the sensing matrix and improve spatial resolution without increasing device size. On the software side, the team reviewed model hyperparameter tuning, compared ground-truth data with predicted outputs, and identified instability in the current PPG signal processing algorithm due to noise and simple threshold-based peak detection. Improvements were proposed to implement a more robust beat-detection algorithm resistant to motion artifacts. Additionally, the app currently generates CSV files containing timestamps, heart rate, and SpO values, but initialization delays and Android permission differences (especially for Android versions above 10) were noted. Issues such as abnormal heart rate readings and possible inconsistencies in skin contact were also discussed, emphasizing the need for further testing of sensor placement.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that edema classification should rely on adaptive personal baselines rather than fixed thresholds, and that implementing hysteresis will improve stability in severity classification. Hardware reliability must be improved by using a stable step-up converter to ensure consistent Arduino operation when powered by the Li-Po battery. Expanding the sensing matrix using a multiplexer remains the preferred approach to increase measurement resolution while maintaining a compact design. On the software side, the existing PPG peak-detection algorithm is not sufficiently robust to noise and motion artifacts and will be replaced with a more reliable method. While CSV logging and visualization in the app are functional, improvements are needed in permission handling, initialization timing, and overall signal stability.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Search for step-up converter, improve heatmap visualization, research more about edema ranges, start drafting conference paper	Raghad Al Agha	Feb. 25
Search for step-up converter, test different peak detection algorithms, research more about state-of-the-art peak detection algorithms, start drafting conference paper	Claritta Khoury	Feb. 25
Model w work on generating random encryption keys for each user and hide them in the apps secrets	Lynn Ariss	Feb. 25
continuation of heatmap generation and hardware implementation, integrating it to the app, and developing a schematic of the full hardware design, working on component layout	Hadi Elham	Feb. 25